SP Corner — Realistic Campus Measurement, Materials & Threshold Guide

Operational Reality:

- **Collection Frequency:** 2 sessions per week (e.g., Tuesday & Thursday, 2 hours each).
- **Monthly Window:** Exactly **8 collection sessions per month** (16 sessions across the 2-month pilot).
- **Campus Context:** Individual plastic generation is modest, but delivery parcel boxes, beverage cans, and food packaging are steady.

1. Comprehensive Campus Recyclables Catalog

To ensure students always have something to bring—even if they do not drink bottled water daily—the desk accepts **four clear campus streams**:

Stream A: Plastics (Rigid & Clean)

- **PET Beverage Bottles:** Mineral water, carbonated soft drinks, juice bottles (rinsed, caps included).
- **HDPE Containers:** Shampoo bottles, body wash, liquid detergent containers from hostel laundry.
- **PP Food & Drink Containers:** Hard plastic takeaway boxes, clean boba/iced coffee cups, instant noodle bowls.
- *Not accepted:* Dirty cling wrap, greasy food packets, straw wrappers.

Stream B: Metals & Aluminium

- **Aluminium Drink Cans:** 100 Plus, Nescafe, Milo, soft drink cans, sparkling water cans.
- **Steel / Tin Cans:** Canned food (tuna, baked beans, condensed milk cans) rinsed clean.
- **Aerosol Cans:** Empty deodorant or air freshener metal canisters.

Stream C: Paper & E-Commerce Packaging (Highest Campus Volume)

- **E-Commerce Cardboard Boxes:** Shopee, Lazada, TikTok Shop, and parcel delivery boxes (flattened).
- **Corrugated Paper Bags & Sleeves:** Delivery parcel envelopes, gift bags.
- **Study & Office Paper:** Printed lecture notes, used exam revision pads, discarded notebooks (metal spirals removed), clean cardboard cartons.
- *Not accepted:* Greasy pizza boxes, used tissue paper, wax-coated drink cartons with liquid inside.

Stream D: Small E-Waste & Broken Appliances

- **Study Gadgets:** Dead power banks, frayed phone cables, broken earphones, spent USB flash drives.
- **Hostel Electronics:** Malfunctioning study lamps, broken electric kettles, travel steam irons, multi-plug extension leads.

2. Realistic Weight Benchmarks: What Does a Student Actually Generate?

Because sessions happen only **twice a week**, a student has only **3 to 4 days of accumulation time** between visits.

Common Campus Item	Individual Weight	What 0.5 kg (500 g) Actually Looks Like	Feasibility Over 3–4 Days
500ml Mineral Water Bottle	\approx 10 - 12\text{ g}	**42 to 50 bottles**	❌ Impossible for 1 person in 3 days
1.5L Big Water Bottle	\approx 30 - 35\text{ g}	**14 to 16 bottles**	⚠️ Difficult unless sharing with roommates
Aluminium Drink Can (320ml)	\approx 14 - 15\text{ g}	**33 to 36 cans**	❌ Impossible for 1 person alone
Shopee / Courier Parcel Box	\approx 120 - 250\text{ g}	**2 to 3 flattened boxes**	✅ Very easy (common delivery habit)
Notebook / Printed Notes Stack	\approx 200 - 300\text{ g}	**1 used notebook or small paper stack**	✅ Highly achievable
Small Electric Kettle / Iron	\approx 400 - 800\text{ g}	**1 broken appliance**	✅ Instantly reaches or beats 0.5 kg

Conclusion for the Developer & Station Volunteers

If the system demands 0.5\text{ kg} of pure plastic per single drop, students will quit in week one. **The rule must be a "Mixed Clean Basket" or a lowered single-drop threshold.**

3. The Calibrated Level 1 Reward Formula

With only **8 collection sessions a month**, a student who attends **3 sessions** is already demonstrating high commitment (attending ~38% of all available campus drop opportunities).

Level 1 Threshold: The "Quick Starter Milestone"

To ensure students earn their first branded token (Sustainability Pioneers pin, sticker, or sustainable pen) within **10 to 14 days** (Sessions 3 or 4 of the month):

- **Requirement Option A (Session Consistency):**
  - Attend **3 verified drop-off sessions** out of the 4 available in the first 2 weeks.
  - Minimum drop per session: **0.15\text{ kg} (150 g)** of mixed recyclables.
- **Requirement Option B (Weight Target):**

- Accumulate a total of **0.60\text{ kg} (600 g) cumulatively** across any 2 to 3 visits.

Visual "Rule of Thumb" for Students (What to bring to pass 150 g):

Any **one** of the following simple bundles qualifies a visit:

- **Option 1:** 1 small Shopee cardboard box + 2 plastic drink bottles.
- **Option 2:** 3 aluminium cans (Milo/100 Plus) + 1 takeaway drink cup.
- **Option 3:** 1 stack of discarded study paper / old notebook.
- **Option 4:** 1 dead electronic device or cable bundle.

4. Complete 8-Session Month Progression (The 3 Levels)

Given the 8-session monthly cadence (16 total sessions in the 2-month pilot):

```
Week 1 (Sessions 1 & 2): Onboarding & First Drops
Week 2 (Sessions 3 & 4): ⭐ LEVEL 1 UNLOCKED (3 Drops or 0.6 kg reached)
Week 3 (Sessions 5 & 6): Mid-Month Consistency Check
Week 4 (Sessions 7 & 8): 🏆 LEVEL 2 UNLOCKED (6 Drops + Top Floor Recycler)
--------------------------- Month 2 ---------------------------
Week 5 to Week 8: Total 12 Drops reached ---> 🎓 LEVEL 3 UNLOCKED (Auto-Certificate + Token)
```

Milestone Level	Target Attendance	Cumulative Weight	Reward Distributed	Fulfillment Window
Level 1: Milestone Starter	**3 Verified Drops** (within Sessions 1–4)	\ge 0.60\text{ kg} total	Branded SP Sticker Pack, Pin, or Eco-Pen	Next session (Session 4 or 5)
Level 2: Monthly Eco-Leader	**6 Verified Drops** (out of 8 in Month 1)	\ge 2.00\text{ kg} total	Featured on App **Hall of Fame**, 1-Week 1.5\times Point Multiplier, Tote Bag	First session of Month 2
Level 3: Certified Pioneer	**12 Verified Drops** (across 16 sessions)	\ge 4.50\text{ kg} total	**Official Digital Certificate** (Auto-PDF) + SP T-Shirt / Token Payout	End of Month 2 Ceremony

5. Developer Logic & Database Parameters

Provide these exact parameters to the developer for backend validation:

```
# System Validation Constants
MIN_SINGLE_DROP_WEIGHT_KG = 0.100  # 100g minimum to register an entry
STAMP_UNIT_WEIGHT_KG = 0.500       # 500g = 1 Stamp
```

```
# Level Thresholds
LEVEL_1_DROPS_REQUIRED = 3
LEVEL_1_MIN_TOTAL_WEIGHT = 0.600   # 600g

LEVEL_2_DROPS_REQUIRED = 6
LEVEL_2_MIN_TOTAL_WEIGHT = 2.000   # 2.0 kg

LEVEL_3_DROPS_REQUIRED = 12
LEVEL_3_MIN_TOTAL_WEIGHT = 4.500   # 4.5 kg

def evaluate_user_tier(user):
    verified_drops = user.transactions.count()
    total_weight = user.cumulative_weight_kg

    if verified_drops >= LEVEL_3_DROPS_REQUIRED and total_weight >=
LEVEL_3_MIN_TOTAL_WEIGHT:
        user.tier = 3
        user.can_generate_certificate = True
    elif verified_drops >= LEVEL_2_DROPS_REQUIRED and total_weight >=
LEVEL_2_MIN_TOTAL_WEIGHT:
        user.tier = 2
        user.hall_of_fame_eligible = True
    elif verified_drops >= LEVEL_1_DROPS_REQUIRED and total_weight >=
LEVEL_1_MIN_TOTAL_WEIGHT:
        user.tier = 1
        user.eligible_for_starter_gift = True

    user.save()
```